# Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

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## Contents

# Abstract

**Background.** Treatment response in aggregated N-of-1 clinical trials reflects three causally distinct mechanisms: pharmacological action of the active compound (BR), the patient's belief that the treatment will work (PB), and natural-history drift in the underlying condition (TV). Standard analysis models lump these into a single 'treatment effect' plus residual error. Whether this lumping biases inference is not unconditional: it depends on the estimand and on whether candidate biomarkers correlate with the non-pharmacological components, a dependence this paper makes precise.

**Methods.** We formalise a three-component decomposition of the within-participant outcome trajectory, each component a modified Gompertz function with participant-specific random effects, and characterise the trial-design contrasts (open-label, blinded discontinuation, crossover) that supply identifiability. We derive the omitted-variable-bias identity for the one-component estimator, which expresses both the lumped treatment effect and the lumped biomarker slope as sums of component-specific terms, and we evaluate the analysis strategies in a pre-registered Monte Carlo study (Study A) calibrated to the prazosin-PTSD reference parameters, comparing a one-component design-contrast analysis with a phase-augmented analysis at 1,000 replicates per alternative cell (5,000 per null cell) across $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 100, 150\}$.

**Results.** The identity shows that the lumped biomarker-by-treatment slope is displaced from the pharmacological slope by a term proportional to the biomarker's covariance with PB and TV, not by the magnitude of those components. Study A confirms this: with a biomarker coupled to BR only, the one-component estimate of the biomarker-by-treatment interaction is essentially unbiased across the whole grid (mean absolute bias 0.016 at $N = 35$ and 0.006 at $N = 150$, within one Monte Carlo standard error of zero), with near-nominal coverage and type I error. The phase-augmented analysis provides no bias reduction and is markedly less powerful (mean power 0.35 versus 0.59 at $N = 35$; 0.62 versus 0.90 at $N = 70$), so it is dominated under this uncontaminated biomarker. The inferential value of decomposition is therefore conditional, not universal.

**Conclusions.** The trial design must supply the drug-on/off and belief contrasts; given those, component-level decomposition is useful when the estimand is the attribution of response to pharmacology versus belief versus natural history, or when a candidate biomarker may correlate with PB or TV. For a biomarker-by-treatment interaction with a biomarker orthogonal to PB and TV, a single-indicator analysis that exploits the design contrast is sufficient and a parametric component model is counterproductive. A pragmatic specification for the cases that warrant it is `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase` with random intercept and AR(1) residual correlation. A coupling-arm study demonstrates the failure mode (the lumped slope is biased in proportion to the biomarker-PB correlation), and Study C shows the bias is recoverable but only under a design that decouples drug exposure from belief: on a balanced-placebo design the belief-covariate decomposition recovers the unbiased pharmacological slope while the one-component estimator does not, whereas on the standard hybrid design no decomposition recovers it. Recovery is therefore conditional on the trial design, not on the analysis alone.

# 1 Introduction

[1]

**Partitioning treatment response into pharmacological ($BR$), placebo-belief ($PB$), and natural-history ($TV$) channels has biological precedent and clinical appeal; this paper rigorously evaluates when the partition is worth making.**

- Precedent runs from pharmacometric disease-progression models [@holford2015; @gomeni2009] through Kaptchuk's empirical decomposition of the placebo response itself [@kaptchuk2008components] to neuroimaging dissociations of drug from expectancy effects [@wager2013nps; @zunhammer2018nps]; each motivates the partition on independent biological grounds.
- Clinical appeal is direct: $BR$ is what persists after expectancy decays; $PB$ drives apparent response in high-placebo-responsiveness populations and misleads precision-medicine stratification when mistaken for pharmacological prediction; $TV$ is what would have changed without any treatment at all.
- Partitioning costs degrees of freedom, requires specific trial-design contrasts for identifiability, and adds analytic complexity; whether the cost is warranted is an empirical and design-dependent question this paper answers.

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Decomposing treatment response into pharmacological, expectancy, and natural-history contributions is not a new idea. Pharmacometrics has long modelled disease progression and drug effect as separable longitudinal processes[1,2]; Kaptchuk and colleagues decomposed the placebo response itself into observation, ritual, and practitioner-relationship components in a three-arm IBS trial[3]; neuroimaging has directly dissociated pharmacological from expectancy-driven analgesia using objective neural signatures[4,5]. The partition has intuitive clinical appeal: a patient whose symptom improvement is pharmacological will benefit from long-term treatment even after novelty fades, whereas one whose improvement is primarily belief-driven may not sustain it as expectations normalise. The question this paper addresses is not whether the partition is biologically coherent – it is – but when it is necessary for valid inference.

[2]

**The optimal choice is conditional: decompose when the candidate biomarker may correlate with $PB$ or $TV$; use the lumped analysis when domain knowledge supports orthogonality.**

- Decompose: COMT val158met genotype as a biomarker in IBS or PTSD trials – COMT regulates dopaminergic expectancy processing and is a documented predictor of placebo-response magnitude [@hall2012comt; @wang2022comt], making it a $PB$ moderator whose lumped biomarker-treatment slope conflates pharmacological and belief-driven prediction. Baseline symptom severity is similarly confounded: it predicts pharmacological response but also couples to regression-to-mean $TV$ [@hrobjartsson2001; @hrobjartsson2010] and heightened treatment expectancy [@kirsch2008; @locher2015].
- Lumped sufficient: CYP metabolizer phenotype as a pharmacokinetic biomarker – enzymatic drug metabolism cannot plausibly couple to belief state or untreated disease trajectory, so the lumped pharmacological slope is already the target estimand. ADRB2 Arg16Gly in asthma bronchodilator trials provides an analogous case: the variant affects receptor downregulation via a direct pharmacological mechanism, lung function is objectively measured by spirometry, and no expectancy pathway links ADRB2 genotype to $PB$.
- In the PTSD/prazosin context of this paper: baseline PCL-5 severity belongs in the decompose column (a psychological measure with documented coupling to both $PB$ and $TV$); an adrenergic receptor variant affecting prazosin binding affinity belongs in the lumped-sufficient column (pharmacological mechanism, no $PB$ coupling).

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The decision criterion follows from an algebraic identity established in section 6: the lumped biomarker-treatment slope decomposes as $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$, where the loadings $w_{PB}$ and $w_{TV}$ are determined by the trial design and the $\beta_{bm}^{C}$ terms are component-specific slopes. Bias is zero if and only if $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$, that is, if and only if the biomarker is uncorrelated with the non-pharmacological components. Whether that condition holds is determined by biology and domain knowledge, not by the magnitude of $PB$ or $TV$ in the study population. A population with a large placebo response but a biomarker orthogonal to it yields an unbiased lumped estimate; a population with a small placebo response but a biomarker that correlates with it yields a biased one.

COMT genotype exemplifies the first category. COMT modulates catechol-O-methyltransferase activity in prefrontal cortex, regulates dopaminergic signalling in expectancy circuits, and predicts placebo-response magnitude in IBS and chronic pain populations[6,7]. A trial using COMT as a predictive biomarker for prazosin response would face a lumped slope that conflates pharmacological prediction with expectancy moderation. CYP2D6 metabolizer phenotype exemplifies the second: the genotype determines hepatic drug metabolism and thereby plasma drug concentration, with no plausible pathway to expectancy or natural-history effects. ADRB2 Arg16Gly in asthma bronchodilator trials provides a further example: spirometric outcomes are objective and receptor binding affinity does not modulate belief. For both, the lumped pharmacological slope is already the target estimand.

[3]

**An algebraic identity establishes the conditional rule exactly; Study A confirms the orthogonal arm; Study B (pre-registered, pilot complete) addresses the coupling arm; together they bound the inferential cost of the choice.**

- The identity (section 6) is exact: bias in the lumped slope $= w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$; it is zero if and only if the biomarker is orthogonal to $PB$ and $TV$, regardless of the magnitude of those components.
- Study A (section 7.3; 1,000 replicates per alternative cell, 5,000 per null) confirms the orthogonal arm: mean absolute bias in the lumped $\hat{\beta}_{bm:D_{bc}}$ is 0.016 against a true value of $-2.25$ at $N = 70$, with no trend across $m_{PB} \in \{0, 1, 3, 6, 10\}$ or $m_{TV} \in \{-1, 0, 1, 2\}$.
- Study B (section 7.4; 100-replicate pilot reported here; production run pre-registered) addresses the coupling arm: pilot data show bias scaling with $c_{bm,PB}$, consistent with the algebraic pre-

diction and motivating the production run.

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The paper proceeds as follows. Section 2 fixes compact notation and states the formal model. Section 3 develops each component with biological motivation, parametric form, and identifiability conditions. Section 4 maps the hybrid open-label-plus-blinded-discontinuation design's phase structure onto the decomposition's identifiability requirements. Section 5 develops the analysis-side methodology, including the phase-augmented LME approximation. Section 6 works through the numerical illustration and derives the algebraic identity that anchors the conditional rule. Section 7 presents Study A (orthogonal arm) and the Study B pilot (coupling arm). Section 8 translates the conditional rule into a practical biomarker-validation decision criterion under the PATH framework[8]. No published N-of-1 trial has applied this evaluation framework; the paper's primary contribution is determining when decomposition changes the inferential answer rather than advocating for it as a universal default.

## 2 Notation and the formal model

[4]

**This section fixes compact notation and states the formal model; a fuller worked-example treatment is in the companion pedagogical document.**

- Definitions are precise enough to support the identifiability and analysis-side arguments in later sections.
- The presentation is kept compact; extended derivations are deferred to the companion document.
- The companion pedagogical document provides numerical worked examples at each formal step.

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In this section we shall fix notation and state the formal model. The three components are defined precisely enough to make the identifiability and analysis-side arguments in later sections rigorous, but we have endeavored to keep the presentation compact. A fuller treatment, with worked numerical examples at each step, is available in the companion pedagogical document at `docs/24-component-decomposition-pedagogy.md`.

[5] where $Y_{it}$ is the observed symptom score, $\mathrm{BL}_i$ is participant $i$'s baseline score, the three components $BR$, $PB$, $TV$ are non-negative

reductions in symptom severity attributable to the three causal mechanisms identified in section 1, and $\varepsilon_{it}$ is observation-level noise. The sign convention is that an increase in any component reduces $Y$; all three components are zero at $t = 0$ by construction (no exposure yet to drug, no belief activation, no elapsed natural history) and grow over the duration of the trial.

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For participant $i$ at trial timepoint $t$:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[\, BR_{it} + PB_{it} + TV_{it} \,\right] \;+\; \varepsilon_{it},$$

[6]

**Each component is driven by its own time-on-state variable, making the components structurally distinct even when they overlap in calendar time.**

- $BR_{it}$ accumulates with time-on-drug; $PB_{it}$ with time-on-belief scaled by expectancy $\eta(\text{phase})$; $TV_{it}$ with calendar time from trial entry.
- The distinct time indices are the structural source of identifiability from phase-contrast data.
- All components are zero at $t = 0$ by construction and grow over the trial duration.

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Each component is a participant-specific function of its own time-on-state variable. $BR_{it}$ depends on time-on-drug, the cumulative exposure to active medication since drug initiation. $PB_{it}$ depends on time-on-belief, modulated by a phase-specific expectancy factor $\eta(\text{phase})$ that captures the patient's confidence that they are receiving active treatment. $TV_{it}$ depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

[7] rescaled so that $C(0) = 0$ and $C(t) \to m_C(1 - e^{-d_C})$ as $t \to \infty$, for $C \in \{BR, PB, TV\}$. The three Gompertz parameters $(m_C, d_C, r_C)$ are participant-specific random effects drawn from population distributions, and the participant-level heterogeneity in these distributions is the principal source of between-person response variability that the framework is designed to characterise.

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Each component follows a modified Gompertz curve:

Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

| Term | Meaning |
|---|---|
| BR | Biological response: pharmacological reduction in symptom score attributable to active drug. |
| PB | Placebo-belief response: reduction attributable to the patient's belief that they are receiving treat |
| TV | Time-variant component: reduction (or increase) attributable to the natural-history trajectory of |
| $\eta$(phase) | Expectancy multiplier for $PB$: 1 in open-label on-drug, $\approx 0.5$ in blinded discontinuation, between |
| $\mathrm{BL}_i$ | Participant $i$'s baseline (pre-treatment) symptom score. |
| $D_{it}$ | Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug. |
| $t_{sd,it}$ | Time since drug discontinuation (off-drug only). |
| $\lambda$ | Carryover decay rate, with half-life $t_{1/2} = \ln 2/\lambda$. |

$$C(t) \; = \; m_C \, \exp(-d_C \exp(-r_C t)) \, - \, m_C \, \exp(-d_C),$$

[8]

**The residual term has AR(1) structure and the three components are permitted to co-vary through cross-component covariances.**

- Within-participant residuals follow $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$.
- Within-time and across-time cross-component covariances permit participant-level heterogeneity in one component to co-vary with the others.
- This structure accommodates biologically plausible correlations, such as high-$BR$ patients also tending toward high-$PB$.

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$\varepsilon_{it}$ is a within-participant residual term with AR(1) serial-correlation structure $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$. The three components are permitted to be jointly correlated through within-time and across-time cross-component covariances $c_{cf1t}$ and $c_{cfct}$ respectively, so that participant-level heterogeneity in any one component can co-vary with heterogeneity in the others.

The principal psychometric and pharmacological terms used below are summarised in Table 1.

# 3 The three components

[9]

**This section presents each component with its biological motivation, Gompertz parameterisation, and identifiability con-**

**ditions to establish the model as causally interpretable.**

- The three components are presented in turn: $BR$ (biological response), $PB$ (placebo-belief), $TV$ (natural history).
- Each component is grounded in independent biological literature before its parametric form is stated.
- The section's purpose is to demonstrate that the model sum reflects causal structure, not mathematical convenience.

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In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterisation, and the design contrasts that identify it. The treatment is organised so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

## 3.1 Biological response (BR)

[10]

**$BR$ is the pharmacological target of regulatory approval and biomarker stratification, independent of belief or natural history.**

- $BR$ arises from chemical action on the body and is zero for inert pills regardless of patient expectation.
- It is the estimand that approval decisions, dose selection, and biomarker-guided stratification require.
- Separating $BR$ from $PB$ and $TV$ is the central precision-medicine problem the framework addresses.

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$BR$ is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief, suggestion, and natural history: a participant who somehow received active drug without knowing it would still develop a measurable $BR$, and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, $BR$ is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around $BR$.

[11]

**PK/PD considerations produce a saturating $BR$ trajectory with a slow initial rise, a peak-growth phase, and a sharp**

**drop to zero on discontinuation.**

- Plasma steady state is reached within a few half-lives; pharmacodynamic integration (e.g. receptor adaptation) adds additional lag.
- For prazosin-PTSD, Raskind et al. (2013) calibrate the onset to several weeks of sustained exposure.
- The saturating shape motivates the Gompertz over linear or simple exponential alternatives.

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The qualitative shape of the $BR$ trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life; brain concentration follows plasma with a short delay; the downstream pharmacodynamic response (reduction in symptom-relevant neural activity, in the case of prazosin's effect on noradrenergic tone during REM sleep) typically requires several days to weeks of sustained exposure to integrate fully[9]. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life $t_{1/2} = \ln 2/\lambda$ rather than an instantaneous drop. The rate of this off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

[12]

**The three Gompertz parameters $(m_{BR}, r_{BR}, d_{BR})$ capture ceiling, onset rate, and climb shape respectively, calibrated to prazosin-PTSD reference values.**

- $m_{BR}$ is the biological ceiling (maximum reduction at infinite exposure); $r_{BR}$ the onset rate; $d_{BR}$ the displacement shaping the early climb.
- Prazosin-PTSD calibration yields $m_{BR} \approx 11$, $r_{BR} \approx 0.42$/week, $d_{BR} \approx 5$.
- These are participant-specific random effects whose population distributions encode between-person response heterogeneity.

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The modified Gompertz form $BR(t) = m_{BR} \exp(-d_{BR} \exp(-r_{BR} t)) - m_{BR} \exp(-d_{BR})$ captures this profile compactly. The asymptote $m_{BR}(1 - e^{-d_{BR}})$ is the biological ceiling, the largest reduction the drug can produce at infinite exposure; the scale parameter $m_{BR}$ equals that ceiling up to the factor $(1 - e^{-d_{BR}})$, which for the calibrated $d_{BR} \approx 5$ exceeds 0.99. The onset rate $r_{BR}$ controls how rapidly the participant approaches the ceiling: high $r_{BR}$ means a fast onset (effect plateaus in one to two weeks), low $r_{BR}$ a slow climb (six to eight weeks). The displacement $d_{BR}$ shapes the climb: high $d_{BR}$ gives an early-onset curve that rises sharply at first, low $d_{BR}$ a more gradual climb. For PTSD prazosin, calibrated values from the[10] reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at roughly four to five weeks of sustained exposure), and $d_{BR} \approx 5$ (moderately steep early climb).

[13]

**The Gompertz is preferred over linear and simple-exponential alternatives because it captures the asymmetric slow-start saturation profile of chronic-condition pharmacodynamics.**

- Linear $BR = \beta t$ is biologically implausible past the first few weeks due to unbounded growth.
- Exponential $BR = m(1 - e^{-rt})$ lacks the slow-start phase from receptor adaptation and gene-expression changes.
- The Gompertz interpolates between both limits and its three-parameter form avoids overfitting; sensitivity analysis is reported separately in `07-gompertz-evaluation`.

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Two alternative families that do not capture the asymmetric saturation profile are worth noting for contrast. A linear function $BR = \beta t$ climbs without bound and is biologically implausible past a few weeks. An exponential decay $BR = m_{BR}(1 - \exp(-rt))$ is acceptable for fast-onset drugs but lacks the slow-start phase that characterises most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high $d$ limit) and linear-in-$t$ growth (low $d$ limit), and its three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting. The Gompertz is a deliberate choice over the two parametric forms most familiar from pharmacokinetic-pharmacodynamic modelling. The sigmoid $E_{\max}$ (Hill) model[11] describes the dependence of effect on concentration (a dose-response curve), whereas $BR(t)$ here is a time-course of accumulating effect at fixed dosing; and the indirect-response (turnover) models that do describe effect time-courses are specified through differential equations on a hidden response variable rather than in closed form. The modified Gompertz retains the slow-start, saturating shape those models produce while remaining a closed-form, three-parameter mean function that is tractable inside a mixed-effects likelihood. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is reported separately as `07-gompertz-evaluation`. The headline result of that evaluation is reassuring for the present framework: in a 1{,}000-replicate-per-cell simulation that generated data under the logistic, hyperbolic-tangent, and piecewise-linear breakpoint families and analysed them under the standard linear-mixed model, the trajectory family was essentially irrelevant to the power of the biomarker-treatment interaction test (the family-to-family spread fell within two Monte Carlo standard errors of sampling noise) and to type I error (0.025 to 0.036 across null cells, with no family-specific inflation). The conclusions drawn here are therefore not artefacts of the Gompertz form, though the regimes the family cannot represent (cyclical or biphasic trajectories) remain genuine exclusions.

[14]

**$BR$ is identified by the blinded-discontinuation contrast, where drug removal collapses $BR$ to zero while $PB$ remains intact.**

- Within-participant on-drug versus off-drug contrasts are the primary identification mechanism.
- The blinded-discontinuation phase provides the cleanest contrast: $BR$ drops, $PB$ is preserved.
- This contrast does not require the analyst to model $PB$ or $TV$ explicitly to recover $BR$.

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The $BR$ component is identified by within-participant on-drug versus off-drug contrasts, formalised in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, $BR$ decays toward zero as the drug clears while $PB$ remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of $BR$ that does not require the analyst to model $PB$ or $TV$ explicitly. The decay is not instantaneous: with the carryover half-life $t_{1/2} = \ln 2/\lambda$ of the analysis-side decay term, residual $BR$ persists into the early off-drug observations, so the contrast is clean only once the discontinuation window is long relative to $t_{1/2}$, and the drug state is best carried by the continuous-decay indicator $D_{it}$ of Table 1 rather than a hard on/off indicator.

## 3.2 Placebo-belief response (PB)

[15]

**$PB$ is the strict placebo response: improvement that persists when the active drug is silently replaced, provided the patient's belief is unchanged.**

- $PB$ is belief-dependent, not drug-dependent; it would survive a silent switch to placebo under sustained belief.
- It is distinct from $BR$ in causal origin: $BR$ requires chemical action; $PB$ requires only belief.
- The definition is strict and operational, not vague or colloquial.

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$PB$ is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

[16]

**$PB$ reflects neurobiologically real mechanisms triggered by belief, not pharmacology; neuroimaging and genetic evidence confirm it is a structured biological phenomenon.**

- Belief-driven endogenous opioid/dopamine release, HPA-axis modulation, and top-down attentional shifts are all documented mechanisms.
- Wager et al. neurologic pain signature dissociates pharmacological from expectancy contributions via fMRI.
- COMT genotype predicts placebo responsiveness, confirming biological structure rather than measurement artefact.

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It is worth being precise about what $PB$ is and is not. $PB$ is not 'fake' improvement; it is a phenomenon clinicians knowingly exploit, a national survey of US internists and rheumatologists finding roughly half report regular use of placebo treatments[12]. As the introduction sets out, the placebo response is mediated by well-characterised neurobiological pathways[13–17]: belief-driven release of endogenous opioids and dopamine, anticipatory hypothalamic-pituitary-adrenal regulation, and top-down attentional shifts, dissociated from pharmacology by the[4] neurologic pain signature[5] and shown to have genetic correlates[6,7]. What makes these mechanisms 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology.

[17]

**$PB$ is identifiable from $BR$ only when belief and treatment allocation differ, as in the blinded-discontinuation phase.**

- The blinded-discontinuation phase is the canonical contrast: patients cannot tell whether they remain on active drug.
- Belief drops to a partial level ($\eta \approx 0.5$) because the patient's expectation is hedged, not absent.
- Designs without a phase that decouples belief from treatment cannot separately identify $PB$ from $BR$.

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For trial design, the relevant point is that $PB$ is separately identifiable from $BR$ if and only if the trial includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in treatment correspondingly drops to a partial level (multiplier roughly 0.5) because the patient's expectation is hedged.

[18]

**The $\eta = 0.5$ blinded-phase expectancy is a modelling assumption operationalising response expectancy, not a measured quantity, and requires sensitivity analysis.**

- The construct is response expectancy in the sense of Kirsch (1985); to our knowledge the residual expectancy under blinded discontinuation has not been measured directly.
- Open-label placebo trials confirm clinically meaningful $PB$-only responses even under explicit disclosure, bounding the residual response away from zero.
- Sensitivity analyses varying $\eta$ are mandatory before clinical conclusions depend on this single parameter.

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$PB$ also follows a Gompertz curve, scaled by the expectancy factor:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB} \exp(-d_{PB} \exp(-r_{PB} t)) + (\text{offset to start at zero}).$$

Typical expectancy values are $\eta = 1.0$ in open-label on-drug, $\eta \approx 0.5$ in blinded discontinuation, and $\eta$ between 0 and 0.5 in open-label placebo (the residual expectation a patient retains when told the previous active phase was beneficial). The quantity $\eta$ operationalises response expectancy in the sense of[18], the patient's anticipation of benefit that the expectancy literature treats as a determinant of experienced symptom change. The choice of $\eta = 0.5$ for the blinded phase is a modelling assumption rather than a calibrated quantity: to our knowledge the residual expectancy a patient carries when uncertain about allocation has not been measured directly as a fraction of the open-label level, and we adopt 0.5 as a deliberately central placeholder. The available indirect evidence is nonetheless that the residual response is substantial. Open-label placebo trials[19–21] demonstrate that a clinically meaningful response persists even when patients are explicitly told they are receiving an inert intervention, which bounds the residual belief-driven response away from zero. Because the value of $\eta$ is not empirically pinned, sensitivity analyses varying it across its plausible range are mandatory before any conclusion is allowed to depend on this single number.

[19]

**Kaptchuk et al. (2008) provide the structural precedent; the BR-PB-TV framework extends the same decomposition logic to the full treatment response.**

- Kaptchuk decomposed the placebo response itself into observation, ritual, and relationship components empirically.
- The present framework applies analogous decomposition logic at the level of drug, placebo, and natural history.
- Both frameworks reject the 'expectancy lump' model in favour of identifiable additive parts.

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The decomposition framework as proposed here has a structural analogue in the placebo-research literature.[3] decomposed the placebo response itself into observation, ritual, and patient-practitioner relationship components in a randomised trial of acupuncture-style placebo for irritable bowel syndrome, demonstrating empirically that the placebo response is not a single 'expectancy' lump but a sum of identifiable parts. The present BR-PB-TV framework is analogous in spirit but operates at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone.

[20]

**$PB$ variance scales with $\eta$, not just its mean; ignoring this**

**heteroscedastic structure miscalibrates standard errors and downstream biomarker tests.**

- Phase-dependent variance follows $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$.
- Open-label phases have high between-patient $PB$ variance; blinded phases have low variance.
- Ignoring heteroscedasticity deflates standard errors in open-label and inflates them in blinded phases, with consequences for CI coverage and biomarker test calibration.

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A subtle property of $PB$ is that its variance also scales with $\eta$, not just its mean. When $\eta$ is high, there is substantial variability across patients in how strongly the placebo response is expressed; high-suggestibility patients show large $PB$, low-suggestibility patients show essentially none. When $\eta$ is low (blinded), the placebo response is uniformly attenuated and the patient-to-patient variance shrinks accordingly. The model therefore parameterises $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic structure deflates standard errors in the open-label phase and inflates them in the blinded phase, with downstream consequences for confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

### 3.2.1 Additivity, subadditivity, and the BR-PB interaction

*[Extended treatment in supplement S2.]*

[21]

**Additivity is a modelling assumption; published literature documents subadditivity between pharmacological and belief effects in some contexts; the phase-augmented analysis partly absorbs it as a side benefit.**

- The generalised model adds $\gamma \cdot BR_{it} \cdot PB_{it}$; $\gamma < 0$ encodes subadditivity, $\gamma = 0$ recovers the additive form.
- Apparent subadditivity can arise from latent-class structure with class-correlated $BR$ and $PB$; distinguishing true interaction from confounding requires a class-aware analysis or the phase contrasts of section 4.
- Simulation quantifying bias scaling with $\gamma$ is deferred to the production programme (section 7).

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The additive decomposition of section 2 treats the three components as independent contributors. Subadditivity – where the combined $BR + PB$ effect falls below the sum of the components measured separately – is documented in some clinical contexts[3,14,22] and can arise from shared neural substrate, ceiling effects, or belief-dynamics adaptation. A generalised model with $\gamma \cdot BR_{it} \cdot PB_{it}$ is identifiable from the same phase contrasts. Apparent subadditivity can also arise from latent-class structure with class-correlated components rather than a direct $BR$-$PB$ interaction; disentanglement requires a class-aware analysis or the phase contrasts provided here. The phase-augmented analysis of section 5 absorbs subadditivity partly as a side benefit.

### 3.3 Time-variant natural-history component (TV)

[22]

**$TV$ is the untreated counterfactual trajectory; it is participant-specific, structured, and can be positive, negative, or near-zero depending on disease class.**

- $TV$ is the symptom change that would have occurred without treatment, integrated over the trial duration.
- It varies in sign across disease classes and across patients within a class.
- It is independent of treatment status and belief, distinguishing it structurally from $PB$.

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$TV$ is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants $TV$ is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others $TV$ is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

[23]

**$TV$ is a structured participant-specific signal, not noise; its distinguishing feature from $PB$ is independence from treatment status and belief.**

- Unlike $\varepsilon_{it}$, $TV$ is participant-level (slow trajectory) and requires explicit modelling, not absorption into noise.
- Off-drug baselines and follow-up windows are the primary data sources for estimating $TV$.

- $TV$ is modulated by neither treatment assignment nor patient belief, whereas $PB$ is modulated by both.

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Like $PB$, $TV$ is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes $TV$ from $PB$ is its origin: $TV$ is independent of treatment status and of belief about treatment, whereas $PB$ is modulated by both.

### 3.3.1 Natural history: existence, testability, and functional form

[24]

**$TV$ sign varies by disease class; the zero-$TV$ assumption is rarely safe in chronic trials; a linear `week` covariate is insufficient because it enforces population-mean linearity and cannot separate $TV$ from $BR$ without the design contrasts of section 4.**

- Disease class determines $TV$ sign: positive for acute recovery, near-zero for stable chronic, negative for progressive conditions; selection-driven enrolment adds a positive $TV$ component regardless of class due to regression to the mean [@hrobjartsson2001; @hrobjartsson2010].
- Zero-$TV$ is safe only when all four conditions hold: stable disease, short trial, non-severity-peaked recruitment, no ancillary care; in most chronic trials at least one fails.
- A `week` covariate enforces linearity, ignores participant-level heterogeneity, and conflates $BR$ with $TV$; the Gompertz form and the design contrasts of section 4 supply identifiability.

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The shape of $TV$ depends on disease class, trial timescale, and enrolment selection. Acute conditions trend $TV$-positive (intrinsic recovery); chronic stable conditions trend near $TV = 0$ on the population mean but with substantial participant-level variance; progressive conditions trend $TV$-negative. Selection-driven enrolment adds a positive $TV$ component regardless of class, because patients enrol at moments of peak severity and regression to the mean then contributes apparent improvement in early trial weeks[23,24]. The zero-$TV$ assumption is testable: a model including $TV$ returns $\hat{m}_{TV}$ with a variance estimate; lumping $TV$ into $\varepsilon$ is worse, because the participant-level natural-history trajectory is structured and correlated with the on/off-drug schedule, producing autocorrelated within-participant residuals and heteroscedastic between-participant residuals that bias the standard errors on $BR$ and $PB$. A simple `week` covariate is insufficient: it enforces a linear, population-mean trajectory and, without the design contrasts of section 4, absorbs both $BR$ and $TV$ into a single coefficient that identifies neither.

# 4 Identifiability and trial-design contrasts

[25]

**Identifiability is supplied by trial design; each phase of the hybrid design contributes a different combination of components to the observed trajectory.**

- The decomposition is mathematically specified but requires trial-design contrasts to be statistically identified.
- This section characterises the hybrid open-label-plus-blinded-discontinuation design as the primary identifiability vehicle.
- Each of the three phases supplies a distinct mixture of $BR$, $PB$, and $TV$.

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The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterise the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

| Phase | Drug? | Belief? | Components present |
|---|---|---|---|
| Open-label on drug | Yes | Full | $BR + PB(\eta = 1) + TV$ |
| Blinded discontinuation | Hidden | Hedged | $BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$ |
| Open-label crossover | Either | Knows | $BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$ |

[26]

**Three phase-by-allocation contrasts isolate $PB$, $BR$, and $TV$ respectively; the LME model combines them but the inference is driven by design.**

- Open-label versus blinded contrast yields approximately $0.5 \cdot PB$, isolating the belief component conditional on a known blinded-phase $\eta$.
- Blinded on-drug versus blinded placebo yields $BR$ once drug carryover has elapsed (since $PB$ and $TV$ match across arms within the blinded phase).
- Off-drug phases yield $TV$ plus residual off-drug $PB$, with no active $BR$ contribution beyond carryover.

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Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$, which isolates the belief component, but only conditional on a known blinded-phase expectancy $\eta$: from a two-phase design just the product $\eta \cdot m_{PB}$ is identified, and separating $\eta$ from $m_{PB}$ requires the open-label placebo phase. The blinded on-drug versus blinded placebo contrast yields $BR$ once drug carryover has elapsed, since $PB$ and $TV$ match across allocation arms within the blinded phase. All phases off drug yield $TV$ plus any residual off-drug $PB$, with no active $BR$ contribution beyond carryover and $PB$ at the off-drug expectation level. The $\mathbf{1}_{\text{on drug}}$ indicator in the phase table above is shorthand for the continuous-decay drug state $D_{it}$ defined in Table 1.

This conditioning on $\eta$ is the framework's principal identification limitation and deserves emphasis rather than a parenthetical. With a two-phase (open-label and blinded) design the data identify only the product $\eta \cdot m_{PB}$, so $PB$ is recovered up to the assumed blinded-phase expectancy; the working value $\eta \approx 0.5$ is a modelling assumption, not an estimate, and absent a belief or expectancy measurement it is not falsifiable from the outcome data alone. The identified set for $m_{PB}$ is accordingly the interval $\{\widehat{\eta m_{PB}}/\eta : \eta \in (0, 1]\}$, which collapses to a point only when $\eta$ is pinned down. Two design features close the gap: an open-label placebo phase, which supplies a second equation separating $\eta$ from $m_{PB}$, and direct measurement of belief, as in the open-hidden and balanced-placebo designs[25,26] that manipulate or record the patient's treatment expectation. We therefore recommend that any trial relying on the decomposition for a $PB$-versus-$BR$ attribution either include such a phase or collect an expectancy measure, and that analyses report sensitivity of the attribution to $\eta$ across its plausible range.

[27]

**Designs lacking specific phases lose identifiability of the corresponding component; the decomposition machinery is only**

**as useful as the design allows.**

- A pure open-label design cannot separate $BR$ from $PB$ because both are present at full strength throughout.
- A pure blinded design cannot estimate $\eta = 1$ $PB$ because no phase achieves full-strength belief.
- Trial designers should verify each component has at least one phase in which it is uniquely present before committing to the design.

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A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate $BR$ from $PB$, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ $PB$, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

[28]

**Design-contrast identifiability is necessary but not sufficient: the three Gompertz trajectories are collinear, so joint estimation is credible only under specific design and sample-size conditions.**

- The components share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present.
- Joint recovery requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough for $TV$ to separate from a decaying $BR$.
- The recoverable region of the design-and-sample-size space is mapped by the pre-registered identifiability diagnostics, not asserted.

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Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for $BR$, the open-label-versus-blinded contrast for $PB$, and an off-drug window for $TV$), and that off-drug window must be long enough relative to the carryover half-life for $TV$ to separate from a still-decaying $BR$. Where these conditions are weak the components are jointly identified only up to a near-collinear direction, and the symptom of that is the rank-deficiency the pilot of section 7 encountered when the full phase-augmented formula was fitted at $N = 35$. We therefore do not assert that the three components are always recoverable. The pre-registered identifiability diagnostics of section 7 (convergence rates of the component-matched fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the $BR$ estimate to the assumed $\eta$(phase) multipliers) are the means by which the recoverable region of the design-and-sample-size space is to be mapped, and the framework's participant-level claims should be read as conditional on falling inside that region.

# 5 Analysis-side methodology

[29]

**Four constraints make a component-matched analysis impractical at trial-relevant sample sizes; a phase-augmented LME captures the important part of the gap at minimal cost.**

- The question of whether to mirror the DGP in the analysis model is reasonable; the section addresses it directly.
- The four constraints are identifiability, degrees of freedom, convergence fragility, and marginal estimand robustness.
- A phase-by-treatment indicator is the recommended modest extension that captures BR-versus-PB sensitivity without the full parametric cost.

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In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data? The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

## 5.1 Four reasons not to match the DGP

[30]

**The first constraint is identifiability: forcing a three-component model on a design with insufficient contrasts produces unidentified parameters, not component estimates.**

- A pure open-label trial cannot distinguish $BR$ from $PB$; both rise from zero together on treatment initiation.
- Fitting three components to a one-contrast design produces implausibly tight standard errors that misrepresent inferential precision.
- The analyst must audit whether the design supplies the necessary contrasts before specifying a component-matched model.

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The first reason is identifiability. The trial design must supply the contrasts that identify each component, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for $PB$, on-drug versus blinded-placebo for $BR$, off-drug timepoints for $TV$). A pure open-label trial has none of these contrasts; in such a trial $BR$ and $PB$ both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

[31]

**The second constraint is degrees of freedom: a component-matched analysis adds approximately a dozen non-linear parameters, inflating moderation estimand variance at $N \in [30, 150]$.**

- Gompertz $BR$ (3), $PB$ (3 + expectancy multiplier), and $TV$ (3) each require participant-specific random effects on the maxima.
- At trial-relevant $N$, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts.
- Additional parametric structure typically inflates moderation estimand variance more than it reduces bias.

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The second reason is degrees of freedom. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation programme) is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz $BR$ (three parameters), a Gompertz $PB$ (three parameters plus a phase-specific expectancy multiplier), and a Gompertz $TV$ (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

[32]

**The third constraint is convergence fragility: component-matched nonlinear mixed models converge below 90% at trial-relevant $N$, contaminating aggregated analyses.**

- Simultaneous Gompertz $BR$, $PB$, and $TV$ fitting requires `nlme::nlme`, Bayesian hierarchical, or structural-equation approximations.
- Each is more sensitive to starting values, prone to local optima, and likely to fail on individual trial datasets than the LME analogue.
- Non-converging cells at below-90% convergence rates contaminate power and bias estimates in simulation studies.

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The third reason is convergence fragility. Component-matched analyses are non-linear in the parameters and their convergence is problematic in practice. Fitting Gompertz $BR$, $PB$, and $TV$ simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. At trial-relevant $N$, the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

[33]

**The fourth constraint is estimand robustness: the `bm:Dbc` marginal interaction is the right estimand for the biomarker-validation question regardless of the BR-PB split.**

- Regulators want to know whether the biomarker predicts treatment response under trial conditions, not a mechanistically decomposed prediction.
- The marginal `bm:Dbc` interaction captures this question whether the moderation is biological or expectation-mediated.
- Component decomposition adds mechanistic precision but is not required for the core biomarker-validation decision.

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The fourth reason is that the moderation estimand is sufficiently robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

[34]

**The inferential target is stated as an estimand: the population biomarker-by-treatment interaction with explicit intercurrent-event handling, while the pharmacological-component target is more ambitious and inherits the heterogeneity-identifiability caveat.**

- The $bm{:}D_{bc}$ estimand is defined by its population, treatment contrast, endpoint, summary, and intercurrent-event strategy in the ICH E9(R1) sense.
- Dropout is handled by a treatment-policy strategy and residual carryover by the modelled exponential decay.
- Recovering a participant-level pharmacological interaction inherits Senn's caveat that true heterogeneity is separable from within-person variation only under repeated phase contrasts.

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It is worth stating the inferential target of the analysis in the vocabulary of the estimands framework[27], because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-by-treatment interaction, the $bm{:}D_{bc}$ coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarised through the continuous indicator $D_{bc}$, its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modelled exponential decay for residual carryover. The pharmacological-component target $\beta_{bm}^{BR}$ is a more ambitious estimand: it is the biomarker-by-$BR$ interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasised for individual treatment effects[28,29]: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

### 5.2 A small extension that usually pays off: phase indicators

[35]

**The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.**

- `phase` absorbs systematic differences in $PB$ expectancy across trial phases.
- `Dbc:phase` detects whether the drug effect shrinks under blinding, a signature of $PB$-mediation.
- `bm:Dbc:phase` detects whether the biomarker-by-treatment interaction shrinks under blinding, a signature that it is partly biomarker-by-$PB$.

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The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
        + bm:Dbc + bm:Dbc:phase
        + (1 | ptID),  correlation = corCAR1(...)
```

Here `phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in $PB$ expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly $PB$-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by-$PB$ rather than biomarker-by-$BR$.

[36]

**The phase-augmented extension adds at most four parameters yet captures the practically relevant belief-attenuation diagnostic without requiring the Gompertz parametric form.**

- Three to four parameters is well within budget at any realistic trial sample size.
- The extension does not separately estimate $BR$ and $PB$ components but identifies the change in apparent drug effect when belief is hedged.
- For most analyses this is the practically relevant question and is answerable without committing to the Gompertz form.

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This extension adds three to four parameters at most, well within budget at any realistic $N$. It does not separately estimate $BR$ and $PB$ as components, but it does identify the change in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form. Where both the lumped and the phase-augmented analyses are reported, the additional `Dbc:phase` and `bm:Dbc:phase` terms introduce a small family of belief-attenuation tests, and the attribution inference they support should be treated as a pre-specified secondary analysis with its multiplicity acknowledged rather than as a second bite at the primary interaction test.

## 5.3 When to fit the full decomposition

[37]

**The full decomposition merits the effort only when $N$ is in the hundreds, the design is fully phase-balanced, and the scientific question requires the BR-PB mechanistic split.**

- Feasibility conditions: several-hundred $N$, dense timepoints in every phase, mechanistic BR-PB split required, and informative Bayesian priors available.
- In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or `brms` can return participant-specific component estimates.
- Outside that regime the phase-augmented LME is the better tool.

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A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific $BR$, $PB$, and $TV$ component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

[38]

**The DGP-analysis asymmetry is deliberate: simulation uses the rich DGP to characterise how a simpler analysis behaves; the analysis need not mirror the DGP to produce useful inference.**

- Simulation controls the DGP and can specify any structure; inference must pay the identifiability cost of every parameter.
- The DGP is a characterisation tool; the analysis is a finite-data inference tool; they have different optimality criteria.
- DGP-matching is not required and is often counterproductive at trial-relevant sample sizes.

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The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterise how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

[39]

**The exponential-decay carryover indicator is a deliberately parsimonious special case of richer carryover models, and the simulation should benchmark it against a distributed-lag alternative.**

- The continuous indicator $D_{bc}$ decays as $\exp(-\lambda t_{sd})$, a one-parameter form tied to a pharmacokinetic half-life.
- Distributed-lag models, the average-period-treatment-effect esti-

mand, and G-estimation relax the single-rate assumption at additional parametric cost.

- The exponential restriction is an assumption, not a finding, and the pre-registered simulation should include a distributed-lag arm to quantify its cost.

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A word on the carryover parameterisation is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator $D_{bc}$ decays as $\exp(-\lambda t_{sd})$ off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al.[30] model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional parameters; Daza[31] formalises an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and G{"a}rtner et al.[32] benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. The pre-registered simulation programme of section 7 should accordingly include a distributed-lag analysis arm, so that the cost of the exponential-decay restriction is quantified against the richer alternative rather than asserted.

# 6 Worked example: information loss without decomposition

## 6.1 What the lumped analysis estimates

[40]

**The lumped treatment-effect estimate has a probability limit equal to the pharmacological component plus belief and natural-history contributions, so its bias does not vanish as $N$ grows.**

- Writing the reduction as $BR + PB + TV - \varepsilon$ and fitting one drug indicator gives plim $\hat{\beta}_D = m_{BR} + b_{PB} + b_{TV}$.
- $b_{PB}$ and $b_{TV}$ are the belief and natural-history terms that load onto the drug indicator under the chosen contrast; both are generally non-negative for an improving outcome.

- Because the displacement is in the probability limit, a larger one-component trial converges on the same biased value: the failure is identification, not precision.

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Before the numerical illustration it is worth establishing algebraically what the one-component analysis actually estimates, because the qualitative failure modes of section 1 are consequences of an identity rather than empirical tendencies that require simulation to demonstrate. Write the within-participant symptom reduction as $R_{it} = \mathrm{BL}_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$. The one-component analysis fits a single drug indicator $D_{it}$,

$$R_{it} \;=\; \alpha + \beta_D\, D_{it} + g(t) + u_i + e_{it},$$

and reports $\hat{\beta}_D$ as 'the treatment effect'. Taking the on-drug minus off-drug contrast that $\beta_D$ targets, and using that the natural-history component is by construction independent of treatment status so that its on/off difference cancels in a within-participant comparison, the probability limit of the estimator is

$$\operatorname{plim} \hat{\beta}_D \;=\; m_{BR} + b_{PB} + b_{TV},$$

where $b_{PB} = E[PB \mid \mathrm{on}] - E[PB \mid \mathrm{off}]$ is the belief differential between the contrasted phases and $b_{TV}$ collects any natural-history contribution that fails to cancel under the contrast actually used; $b_{TV}$ is zero for a clean on/off within-participant contrast and strictly positive for a baseline-anchored pre-post analysis, in which case $b_{PB} = E[PB]$ and $b_{TV} = E[TV]$. To fix which case is in play below: the one-component analysis used in the worked example and the simulations of this paper is the design-contrast (drug-on/off) form, so $b_{TV} \approx 0$ there and the operative displacement is the belief term $b_{PB}$; the baseline-anchored pre-post form, which additionally carries $b_{TV} = E[TV]$, is the worse case retained only for comparison. Both terms are generally non-negative for an improving outcome, so the lumped estimate over-states the pharmacological component $m_{BR}$. The displacement is a property of the probability limit and is unchanged as $N \to \infty$: a larger one-component trial converges on the same biased value. This is the precise sense in which the placebo-attribution failure of section 1 is a problem of identification, not of precision.

[41]

**The biomarker slope from a lumped analysis decomposes additively into component-specific slopes, so a biomarker cor-**

**related with $PB$ or $TV$ registers as a predictor even with no pharmacological association.**

- By linearity of covariance, $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$.
- A non-zero $\beta_{bm}^{PB}$ or $\beta_{bm}^{TV}$ produces a non-zero lumped slope even when the pharmacological slope $\beta_{bm}^{BR}$ is exactly zero.
- The result is an algebraic identity under the additive decomposition, not a sample-size-dependent tendency.

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The same algebra settles the biomarker-validation failure. The precision-medicine analysis regresses a participant-level response summary on a baseline biomarker $B_i$. If that summary is the lumped per-participant effect, which estimates $m_{BR,i} + w_{PB}\, m_{PB,i} + w_{TV}\, m_{TV,i}$ for design-determined loadings $w_{PB}, w_{TV} \geq 0$, then by the linearity of covariance the fitted biomarker slope has probability limit

$$\beta_{bm}^{\text{lumped}} = \frac{\operatorname{Cov}(m_{BR} + w_{PB}m_{PB} + w_{TV}m_{TV},\, B)}{\operatorname{Var}(B)} = \beta_{bm}^{BR} + w_{PB}\,\beta_{bm}^{PB} + w_{TV}\,\beta_{bm}^{TV},$$

where $\beta_{bm}^{C} = \operatorname{Cov}(m_C, B)/\operatorname{Var}(B)$ is the component-specific slope. A biomarker correlated with placebo responsiveness or with the natural-history trajectory, so that $\beta_{bm}^{PB}$ or $\beta_{bm}^{TV}$ is non-zero, therefore registers a non-zero lumped slope even when its pharmacological association $\beta_{bm}^{BR}$ is exactly zero. This is the biomarker-validation failure of section 1, and it too is an identity under the additive decomposition rather than an empirical tendency: no sample size separates the lumped slope from a genuine pharmacological one, because the two coincide in the estimand whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$.

We make two qualifications explicit, because they bound how much weight the framework can carry. First, both displays are immediate corollaries of the additive decomposition and the linearity of the projection that defines the estimator; they are not a derived result in any deeper sense, and their value is in naming the displacement terms, not in the algebra. Second, they assume the components combine additively. Under the subadditive interaction documented in section 3.3, where the joint effect of $BR$ and $PB$ falls below their sum, a first-order correction proportional to $\operatorname{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the biomarker slope, and the identities above should be read as the leading behaviour in the additive case rather than as exact under interaction. The empirical magnitude of that correction term is itself an open question that the framework's $\gamma$-interaction simulation (section 3.3) is designed to address.

[42]

**The bias displaces the estimand rather than inflating variance, so a significance test of the lumped coefficient cannot detect it; the decomposition is what locates it.**

- The lumped coefficient is displaced from $m_{BR}$ by $b_{PB} + b_{TV}$ but remains a well-estimated quantity with a valid Wald test of its own nullity.
- Detecting the bias requires comparing $\hat{\beta}_D$ to $m_{BR}$, a comparison only the decomposition or an isolating phase contrast supplies.
- This is why a rejection-rate summary is the wrong instrument, and accounts for the flat rejection rates across the placebo-strength axis in the section 7.1 pilot.

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Two consequences for how the framework should be evaluated follow. First, the quantity that carries the bias is the location of the estimand, plim $\hat{\beta}_D$, not its sampling variability: the lumped coefficient remains a well-estimated number with a valid Wald test of its own nullity, and that test rejects at the rate dictated by the displaced value. A significance test of the lumped coefficient therefore cannot, in principle, detect the bias, because the bias does not move the coefficient toward zero. Detecting it requires comparing $\hat{\beta}_D$ against $m_{BR}$, which is exactly the comparison the decomposition, or a phase contrast that isolates $m_{BR}$, makes available and the lumped analysis does not. Second, and as a corollary for the simulation evidence, a rejection-rate or power summary is the wrong instrument for this bias: the right target is the coefficient itself and its offset from $m_{BR}$. This is why the pilot of section 7.1 found the biomarker-by-treatment rejection rate essentially flat across the placebo-strength axis even though that axis is precisely what drives the predicted bias; the prediction lives in the mean of $\hat{\beta}_{bm:D_{bc}}$, and a production run that means to test it must read the bias off the coefficient, not off the test.

## 6.2 A numerical illustration

[43]

**Figure 1 illustrates the diagnostic contrast: two participants with identical open-label responses but opposite $BR$-$PB$ profiles become distinguishable only in the blinded-discontinuation phase.**

- Participant A ($m_{BR} = 4$, $m_{PB} = 6$) and Participant B ($m_{BR} = 8$, $m_{PB} = 1$) both show approximately ten points of open-label improvement, indistinguishable to a one-component analysis that estimates drug at 7.75 points.

- In the blinded-discontinuation phase A retains four points ($PB$ persists at $\eta = 0.5$) while B collapses to 1.5 points (drug cleared, minimal $PB$): a seven-point diagnostic contrast.
- The three-component decomposition recovers $\hat{m}^A_{BR} = 4$, $\hat{m}^B_{BR} = 8$, $\hat{m}^A_{PB} = 6$, $\hat{m}^B_{PB} = 1$ – six clinically actionable quantities versus one biased average; predictive-biomarker validation requires these.

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*[Figure 1 here: two-panel trajectory plot. Left panel shows open-label phase (weeks 1–8) with overlapping response trajectories for Participants A and B. Right panel shows blinded-discontinuation phase (weeks 9–12) with the diagnostic divergence: A retains four points while B collapses to 1.5. Component contributions are labelled in each panel.]*

Two participants in a sixteen-week N-of-1 trial of prazosin for PTSD share the same baseline but differ in latent traits: Participant A has $m^A_{BR} = 4$, $m^A_{PB} = 6$; Participant B has $m^B_{BR} = 8$, $m^B_{PB} = 1$. Both reach approximately ten points of open-label improvement by week eight, making them indistinguishable to a one-component analysis that estimates a population drug effect of 7.75 points. The blinded-discontinuation phase supplies the identifying contrast: A retains four points of improvement (belief persists at $\eta = 0.5$) while B returns nearly to baseline (drug cleared, minimal belief). The three-component decomposition recovers all six component estimates by exploiting this phase differential; the one-component model cannot.

# 7 Simulation study

[44]

**The simulation study follows ADEMP and quantifies bias and identifiability properties of three analysis strategies across a prazosin-PTSD-calibrated parameter grid.**

- The Morris (2019) ADEMP framework structures the study: Aims, DGP, Estimands, Methods, Performance measures.
- The pre-registration is at `analysis/scripts/component-decomposition/00-ademp-pre-registrat`
- Three analysis strategies (one-component, phase-augmented, full decomposition) are compared across the parameter grid.

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The simulation study reported and pre-registered in this section is structured under the[33] ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The pre-registered plan is at `analysis/scripts/component-decomposition/00-ademp-pre-registration.md`. A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in section 5 across a parameter grid.

## 7.1 Pilot validation

*[Details in supplement S1.]*

[45]

**A 100-replicate pilot confirmed the analysis pipeline is operational; two design-level findings inform the production specification.**

- Convergence was 1.0 across six cells at $N = 35$; per-replicate output is checkpointed at `analysis/data/quick-sim/`.
- At $N = 35$, rank-deficiency dropped the $D_{bc}$:phase interaction term, requiring larger $N$ and a richer design in the production programme.
- The predicted bias manifests in the mean of $\hat{\beta}_{bm:D_{bc}}$, not the Wald-test rejection rate; the production programme reports coefficient means against true values.

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A 100-replicate pilot at $N = 35$ on a 6-cell grid confirmed the end-to-end pipeline. Two observations inform the production specification: at $N = 35$, rank-deficiency dropped the $D_{bc}$:phase interaction term, removing the key contrast; and the predicted bias appears in the coefficient mean, not the rejection rate, so the production programme reports $\hat{\beta}_{bm:D_{bc}}$ against its implied true value rather than power alone. Full details and exploratory summaries are in supplement S1.

## 7.2 Production-simulation specification

[46]

**The production programme follows the ADEMP framework with four deliverables; the realised Study A run in section 7.3 is the first of these.**

- The deliverables are interaction-estimate bias, full-decomposition identifiability, type I error and power, and participant-level signal recovery.

- Sample sizes $N \in \{35, 70, 100, 150\}$ cross a *pb_strength* grid; type I cells target MCSE 0.003 at 5,000 replicates.
- Driver scripts, checkpointed outputs, and the ADEMP pre-registration reside under `analysis/scripts/component-decomposition/`.

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The production programme follows the[33] ADEMP framework and targets four deliverables: the bias of $\hat{\beta}_{bm:D_{bc}}$ for the one-component, phase-augmented, and full-decomposition analyses across $N \in \{35, 70, 100, 150\}$ and a *pb_strength* grid; identifiability metrics for the full-decomposition fit (convergence, variance-inflation factors, and sensitivity of the $BR$ estimate to the $\eta$(phase) multipliers); type I error and power for the interaction test under an $\eta$-mismatch, $TV$-magnitude, and dropout sensitivity grid, with null cells at $n_{\text{reps}} = 5{,}000$ for MCSE 0.003; and the participant-level interaction signal recovered as a function of $(m_{BR}, m_{PB}, m_{TV})$, which the lumped-total regression does not preserve. The `tidyverse` implementation collection drives the grid; driver scripts, checkpointed cell-level summaries, per-replicate output, and the committed ADEMP pre-registration all reside under `analysis/scripts/component-decomposition/`. The realised Study A run reported next used 1,000 replicates per alternative cell.

## 7.3 Study A: orthogonal-biomarker arm (1,000 replicates per alternative cell)

[47]

**Study A is the orthogonal-biomarker arm of the conditional: with $c_{bm,PB} = 0$, the one-component estimate is essentially unbiased across the entire $(m_{PB}, m_{TV})$ grid, exactly as the algebraic identity requires.**

- The true biomarker-by-$BR$ effect is $-2.25$; mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one Monte Carlo SE of zero.
- Bias shows no monotone trend across $m_{PB} \in \{0, 1, 3, 6, 10\}$ or $m_{TV} \in \{-1, 0, 1, 2\}$; it behaves as sampling noise around zero.
- This is the predicted consequence of the identity in section 6.1: the DGP couples the biomarker to $BR$ only, so $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$ and the lumped slope is unbiased by construction.

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The production Study A run (1,000 replicates per alternative cell, 5,000 per null cell, 80 alternative cells crossing $m_{PB} \in \{0, 1, 3, 6, 10\}$,

$m_{TV} \in \{-1, 0, 1, 2\}$, and $N \in \{35, 70, 100, 150\}$, plus four null cells) is the orthogonal-biomarker arm of the conditional evaluation. The data-generating process couples the baseline biomarker to the $BR$ component only, through the on-drug biomarker-$BR$ covariance channel, so the true biomarker-by-treatment effect is the biomarker-by-$BR$ slope, $\beta_{bm:D_{bc}} = -c_{bm}\,\sigma_{BR}/\sigma_{bm} = -2.25$, and the biomarker is by construction uncorrelated with $PB$ and $TV$. Under the one-component analysis the mean estimated $\hat{\beta}_{bm:D_{bc}}$ tracks this value across the whole grid: the mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one Monte Carlo standard error (respectively about 0.028, 0.019, 0.009, and 0.007) of zero, with no monotone trend across $m_{PB}$ or $m_{TV}$. The predicted placebo-attribution bias of the biomarker-by-treatment estimate does not appear.

This is not a failure of the simulation; it is the behaviour the identity of section 6.1 requires. That identity writes the lumped biomarker slope as $\beta^{BR}_{bm} + w_{PB}\beta^{PB}_{bm} + w_{TV}\beta^{TV}_{bm}$. With the biomarker coupled to $BR$ alone, $\beta^{PB}_{bm} = \beta^{TV}_{bm} = 0$, so the lumped slope equals the pharmacological slope no matter how large $m_{PB}$ or $m_{TV}$ grows. Study A varies the magnitudes of the unmodelled components, which the identity shows to be the innocuous axis; the axis that produces biomarker-validation bias is the biomarker's correlation with $PB$ or $TV$, which this DGP holds at zero. The clean null is therefore a confirmation of the identity and a correction to the looser claim of section 1 that large $PB$ or $TV$ by itself biases biomarker validation. It does not; biomarker-component coupling does.

[48]

**The phase-augmented analysis provides no bias reduction and incurs a substantial power penalty, so it is dominated under this uncontaminated DGP.**

- Mean power at $N = 35$ is 0.35 (phase-augmented) versus 0.59 (one-component); at $N = 70$, 0.62 versus 0.90; both reach approximately 1.0 by $N = 100$.
- Phase-augmented mean absolute bias is no smaller than one-component at any $N$ (0.028 versus 0.016 at $N = 35$).
- Convergence is approximately 100% for both analyses at every $N$ (minimum 0.998), so the rank-deficiency that degraded the $N = 35$ pilot does not recur here.

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The comparison between analyses is equally clear. The phase-augmented analysis is not less biased than the one-component analysis at any sample size (its mean absolute bias is 0.028 versus

0.016 at $N = 35$ and is comparable elsewhere), and it is markedly less powerful for the biomarker-by-treatment test: mean power across the grid is 0.35 versus 0.59 at $N = 35$ and 0.62 versus 0.90 at $N = 70$, with both analyses reaching essentially full power by $N = 100$. Because the DGP contains no biomarker-$PB$ contamination for the phase contrast to remove, the extra phase-by-treatment terms add estimation cost without buying any bias reduction, and the analysis is dominated. Convergence was approximately complete for both analyses at every sample size (minimum convergence rate 0.998), so the rank-deficiency that forced dropping the `Dbc:phase` term in the $N = 35$ pilot, which we had advanced as the explanation for the pilot's inverted power ranking, is not operative in the production run: the inverted ranking persists with the full formula fitted and converged. The pilot's power inversion was therefore a genuine feature of this DGP, not an artifact of rank-deficiency.

Calibration is adequate for both analyses. Nominal-95% confidence-interval coverage of $\beta_{bm:D_{bc}}$ ranges from 0.96 to 0.99 for the one-component analysis (slightly conservative) and from 0.94 to 0.98 for the phase-augmented analysis. Type I error at the four null cells is controlled, ranging from 0.031 to 0.056 for the one-component analysis and 0.038 to 0.049 for the phase-augmented analysis; the one-component estimate at $N = 100$ (0.056, MCSE 0.003) sits 1.9 standard errors above nominal but is within the simulation's Monte Carlo uncertainty.

[49]

**Study A confirms the identity but cannot exhibit the biomarker-validation failure; a contaminated-biomarker pilot reported in the next subsection provides the decisive evidence.**

- As pre-registered, Study A varies component magnitude with the biomarker fixed to $BR$, so it probes an axis the identity predicts to be null.
- The decisive axis is $c_{bm,PB}$, the biomarker-$PB$ correlation; this is varied in the contaminated-biomarker pilot below.
- On current evidence the phase-augmented analysis is not recommended for the uncontaminated-biomarker regime; the pilot below evaluates it under contamination.

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Two conclusions follow for the Study A programme. First, the result sharpens rather than supports the section 1 framing: the biomarker-validation failure is real as an identity, but it is driven by the biomarker's coupling to $PB$ or $TV$, not by the magnitude of those components, and Study A as pre-registered varies the latter while holding the former at zero. The decisive axis is the biomarker-$PB$ correlation $c_{bm,PB}$, which the contaminated-biomarker pilot in the following subsection varies directly. Second, on the present evidence the phase-augmented analysis carries a real power cost without a compensating bias reduction under an uncontaminated biomarker, so it should not be recommended as a default for this design; the pilot below evaluates whether contamination changes this verdict.

### 7.4 Study B pilot: coupling-arm simulation (100 replicates per cell)

[50]

**A pilot varying the biomarker-$PB$ correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ confirms that bias in the one-component estimator scales with coupling strength, not component magnitude, and that the phase-augmented analysis provides no protection.**

- At $N = 150$, mean $\hat{\beta}_{bm:D_{bc}}$ shifts from $-2.25$ (true) to $-1.74$ at $c_{bm,PB} = 0.45$ (bias +0.51, 30 Monte Carlo SEs from zero), confirming the identity's prediction.
- The bias is driven by $c_{bm,PB} \times \sigma_{PB}$, not by the population-mean $m_{PB}$: cells with $m_{PB} = 0$ and $m_{PB} = 6$ show identical bias at every $c_{bm,PB}$ level, because the individual-level $PB$ variance ($\sigma_{PB} = 2$) provides the contamination channel even when the mean is zero.
- The phase-augmented analysis shows the same bias as the one-component analysis at every contamination level, offering no protection against biomarker-$PB$ coupling.
- This pilot uses 100 replicates per cell; a production run at 1,000 replicates per cell is required to confirm the pattern and assess type I error under contamination.

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A supplementary pilot varying the biomarker-$PB$ correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ at two $PB$ magnitude levels ($m_{PB} \in \{0, 6\}$) and two sample sizes ($N \in \{70, 150\}$), using 100 replicates per cell, confirms the identity-based prediction and reveals an unexpected structural result.

At $N = 150$ the one-component bias scales monotonically with $c_{bm,PB}$: mean $\hat{\beta}_{bm:D_{bc}}$ moves from $-2.31$ ($c_{bm,PB} = 0$, bias $-0.06$) to $-2.37$ ($c_{bm,PB} = 0.15$, bias $-0.12$) to $-2.08$ ($c_{bm,PB} = 0.30$, bias $+0.17$) to $-1.74$ ($c_{bm,PB} = 0.45$, bias $+0.51$). The last value sits 30 Monte Carlo standard errors from the true $-2.25$, establishing the effect unambiguously. The direction of bias is positive (toward zero): high-$c_{bm,PB}$ participants experience stronger $PB$, which in the open-label phase adds to the apparent drug effect, but this elevation is partially absorbed by the $bm$ main effect, leaving the $bm : D_{bc}$ interaction coefficient attenuated relative to the pure pharmacological value.

The unexpected result is that $m_{PB}$ does not modulate the bias: cells with $m_{PB} = 0$ and $m_{PB} = 6$ produce identical bias at every contamination level (differences $\leq 0.006$ at $N = 150$, well within Monte Carlo error). The driver is $c_{bm,PB} \times \sigma_{PB}$, not $c_{bm,PB} \times m_{PB}$. Even when the population-mean $PB$ amplitude is zero, individual participants have nonzero $PB$ draws (standard deviation $\sigma_{PB} = 2$), and coupling the biomarker to that variance is sufficient to contaminate the estimator. A true no-$PB$ control requires $\sigma_{PB} = 0$, not $m_{PB} = 0$. This refines the Study A design: varying $m_{PB}$ at $c_{bm,PB} = 0$ correctly produced zero bias, but because the relevant null was $c_{bm,PB} = 0$, not $m_{PB} = 0$.

The phase-augmented analysis provides no protection: its bias at every $(c_{bm,PB}, m_{PB}, N)$ cell is indistinguishable from the one-component bias. Adding phase-by-treatment interactions to the model does not decompose the contamination because both analyses see the biomarker-$PB$ covariance through the same open-label versus blinded-discontinuation contrast; the contrast changes the weight on $PB$ (via expectancy) but does not isolate the $PB$ component's contribution to the biomarker slope. Separating that contribution requires both a design that decouples drug exposure from belief and a decomposition that exploits it; Study C shows that the standard hybrid design cannot achieve this separation, whereas a balanced-placebo design can.

The $N = 70$ results are uninformative at 100 replicates (Monte Carlo standard error $\approx 0.06$, comparable to or larger than the bias at moderate $c_{bm,PB}$). The production run will use 1,000 replicates per cell to establish the $N = 70$ pattern and 5,000 null replicates to assess type I error under contamination. Until the production run is complete, the quantitative bias estimates from this pilot should be treated as directionally informative but imprecise.

[51]

**The contaminated-biomarker pilot reframes the paper's practical guidance: the risk of biomarker-validation failure arises from biomarker-$PB$ correlation in the study population, not from the presence of strong placebo response per se.**

- A trial in a population where the biomarker is uncorrelated with placebo responsiveness can use the one-component estimator without bias, regardless of how large the placebo response is.

- A trial where the biomarker covaries with optimism, suggestibility, or treatment-seeking history is at material risk of contamination; bias of +0.51 (23% of the true effect) at $c_{bm,PB} = 0.45$ illustrates the practical scale.
- Neither the one-component nor the phase-augmented analysis removes this bias; full component decomposition is required.

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The pilot results reframe the paper's practical guidance. The risk of biomarker-validation failure in an aggregated N-of-1 study arises not from the presence of a strong placebo response in the population but from the biomarker's correlation with individual-level placebo responsiveness. In a population where the baseline biomarker is uncorrelated with each participant's placebo-belief response amplitude, the one-component estimator recovers the pharmacological interaction without bias, and the full decomposition machinery is not required. In a population where the biomarker covaries with psychological characteristics that predict placebo response – optimism, prior treatment experience, suggestibility, expectancy at enrolment – the one-component analysis conflates pharmacological and placebo-belief prediction, and the scale of that conflation grows with the coupling strength. At $c_{bm,PB} = 0.45$, a value that equals the pharmacological coupling used throughout this paper, the one-component bias is approximately 0.51 points on the interaction scale, representing 23% of the true pharmacological effect. This is a practically meaningful distortion for any precision-medicine stratification decision.

The production run of the contaminated-biomarker study, at 1,000 alternative-cell replicates and 5,000 null-cell replicates, is required to confirm these estimates, establish the $N = 70$ pattern, and characterise type I error under contamination. Results are pending.

## 7.5 Study C: recovery under a belief-decoupling design

[52]

**Study C shows the contamination is recoverable, but only under a design that decouples drug exposure from belief: a belief-covariate decomposition on a balanced-placebo design returns the unbiased pharmacological slope across the feasible contamination range, while the one-component estimator does not.**

- In the MVN process the biomarker-$BR$ coupling loads on the drug state $D_{bc}$ and the biomarker-$PB$ coupling scales with the expectancy $\eta$; the decomposition $Sx \sim bm+t+D_{bc}+\eta+bm{:}D_{bc}+$

$bm{:}\eta$ isolates $\beta^{BR}_{bm}$ on $bm{:}D_{bc}$ only if $D_{bc}$ and $\eta$ vary independently.
- On the standard hybrid design they are collinear (analysis-stage correlation $\approx +0.6$), and the belief-covariate, blinded-stratum, and Gompertz-basis decompositions all fail there, no better than the one-component analysis.
- A balanced-placebo design adds a covert on-drug phase ($\eta = 0$) and an open-placebo phase ($\eta = 1$), decoupling the contrasts ($\approx -0.66$) and enabling recovery.

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Study C asks whether the contamination demonstrated in the coupling-arm study can be removed by decomposition, and finds that the answer depends on the trial design. In the MVN process the biomarker-$BR$ coupling acts only on drug and so loads on the drug-state contrast $D_{bc}$, whereas the biomarker-$PB$ coupling scales with the expectancy $\eta$; interacting the biomarker with both isolates $\beta^{BR}_{bm}$ on $bm{:}D_{bc}$, but only where $D_{bc}$ and $\eta$ vary independently. On the standard hybrid design they do not (on-drug time and belief accumulate together, with an analysis-stage correlation of about $+0.6$), and we verified that the belief-covariate decomposition, a blinded-stratum drug contrast, and a Gompertz-basis component decomposition all fail to recover $\beta^{BR}_{bm}$ there. The remedy is a design that breaks the collinearity: a balanced-placebo (open-hidden) design[25,26] interposes a covert on-drug phase, in which the drug is given without the participant's knowledge ($\eta = 0$), and an open-placebo phase, in which the participant is off drug but believes treatment continues ($\eta = 1$), reducing the $D_{bc}$-$\eta$ correlation to about $-0.66$. \end{orig}

[53]

**At 1,000 replicates per cell the decomposition is unbiased across the contamination range while the one-component estimator degrades monotonically; recovery is conditional on the design and bounded by positive-definiteness.**

- At $N = 150$ the decomposition bias of $\hat{\beta}_{bm:D_{bc}}$ stays within one Monte Carlo SE of zero across $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ ($-0.022$, $+0.008$, $+0.021$, $-0.026$; MCSE $\approx 0.02$), while the one-component bias rises to $+0.16$, $+0.34$, $+0.48$.
- Interval coverage is near-nominal for the decomposition (0.94–0.96) and falls to 0.86 for the one-component analysis at the highest contamination; convergence is 100 percent.

- The grid is capped at $c_{bm,PB} = 0.30$ because beyond approximately 0.45 the implied covariance is non-positive-definite and the contaminated DGP is no longer faithfully representable.

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The production run (1{,}000 replicates per cell, $N \in \{70, 150\}$, $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$) confirms the recovery. At $N = 150$ the decomposition tracks the pharmacological target $-2.25$ across the contamination range, with bias $-0.022$, $+0.008$, $+0.021$, and $-0.026$, every value within one Monte Carlo standard error (about 0.02) of zero, while the one-component estimator develops a monotone bias of $+0.16$, $+0.34$, and $+0.48$ as $c_{bm,PB}$ grows. Interval coverage holds near nominal for the decomposition (0.94 to 0.96) and degrades to 0.86 for the one-component analysis; the $N = 70$ cells show the same pattern at lower power. Recovery is therefore conditional on the design, not on the analysis alone: the decomposition is necessary but a design that varies belief independently of drug exposure is the enabling condition. The practical recommendation is that a biomarker-validation trial in which the candidate biomarker may correlate with placebo responsiveness should be fielded as a balanced-placebo or open-hidden design rather than the standard hybrid design. Per-replicate output is at `analysis/data/derived_data/component-decomposition/study-b-balanced/`. Recovery beyond the positive-definite range, participant-level component recovery, and replication under Architecture A coupling remain open. \end{orig}

# 8 Application to predictive-biomarker validation

[54]

**The precision-medicine question – whether a biomarker predicts pharmacological response – is answered by the lumped total only when the biomarker is uncorrelated with $PB$ and $TV$; otherwise it requires decomposing the response.**

- The question is whether the biomarker predicts the $BR$ component, not the lumped total.
- The PATH statement formalises this as the tar-

get estimand for predictive biomarker validation.
- The BR-PB-TV decomposition is the mechanism that separates the pharmacological signal when biomarker-component coupling cannot be ruled out.

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The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalised in the PATH statement[8]. Whether it depends on the BR-PB-TV decomposition is itself conditional: by the identity of section 6.1, the lumped biomarker slope equals the pharmacological slope when the biomarker is uncorrelated with $PB$ and $TV$, and departs from it only by the biomarker's covariance with those components. The decomposition is therefore the mechanism the question requires precisely in the regime where that covariance is non-negligible.

[55]

**A lumped-total biomarker regression conflates pharmacological and placebo-belief prediction, producing a stratification biomarker that mis-identifies responders.**

- $\hat{\beta}_{bm}$ from the lumped regression contains both pharmacological and placebo-belief prediction in unknown proportions.
- If $PB$ covaries with the biomarker distribution, flagged 'high responders' include high-$PB$ responders who will not benefit pharmacologically.
- Prospective application of such a biomarker would systematically mis-stratify patients for prescription decisions.

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A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the lumped total contains. If $PB$ varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then $\hat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-$BR$ responders and high-$PB$ responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

[56]

**A valid pharmacological predictor requires a non-zero $\hat{\beta}_{bm}^{BR}$; significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ indicate a psychological or prognostic biomarker, not a pharmacological one.**

- The sole criterion for pharmacological prediction is a meaningfully non-zero $BR$-component regression coefficient.
- A biomarker correlating only with $PB$ selects placebo responders, not drug responders.
- A biomarker correlating only with $TV$ is prognostic but not predictive in the pharmacological sense.

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The right approach is to regress each component on the biomarker:

$$
\begin{aligned}
BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\
PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\
TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}.
\end{aligned}
$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with $PB$ would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with $TV$ would be a prognostic marker, not a predictive one.

[57]

**When biomarker-component coupling cannot be ruled out, the decomposition is the mechanism by which the precision-medicine question is posed; when it can, the lumped analysis already answers it.**

- Where the biomarker may correlate with $PB$ or $TV$, separating $BR$ from them is what prevents validation from conflating pharmacological and non-pharmacological prediction.
- The phase-augmented `bm:Dbc:phase` interaction is a small-$N$ proxy, but Study A shows it buys nothing under an uncontaminated biomarker, so it is warranted only when contamination is plausible.
- The full decomposition recovers component-specific regressions explicitly when sample size permits.

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The decomposition is therefore the mechanism by which the precision-medicine question is posed whenever the candidate biomarker may correlate with $PB$ or $TV$. Where the biomarker can be assumed orthogonal to the non-pharmacological components, the identity of section 6.1 shows the lumped analysis already targets the biomarker-by-$BR$ estimand, and the simulation of section 7.3 confirms it recovers that estimand without bias. The phase-augmented linear-mixed analysis introduced in section 5 retains a useful subset of the decomposition machinery at small $N$ (the `bm:Dbc:phase` interaction tests whether the biomarker-by-treatment effect attenuates under blinding, the practical proxy for biomarker-by-$PB$ contamination), but Study A shows this proxy earns its degrees of freedom only when such contamination is present: under an uncontaminated biomarker it reduced no bias and lost power. The full decomposition recovers the component-specific regressions explicitly when $N$ permits and when the scientific question warrants the cost.

# 9 Discussion

[58]

**The three-component framework reframes treatment response as a structured sum of causally distinct contributions, and the present work maps when acting on that decomposition actually changes inference.**

- The framework specifies which trial-design contrasts identify which components.
- Acting on the decomposition is valuable conditionally: for attribution estimands, and for biomarkers that may correlate with $PB$ or $TV$, but not for a biomarker-by-treatment interaction with a biomarker orthogonal to them.
- The target estimand for biomarker validation is the biomarker-by-$BR$ association; whether a lumped analysis already recovers it depends on the biomarker's coupling to $PB$ and $TV$.

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The three-component decomposition is the formal expression of the claim that 'treatment response' in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. What the present work establishes, analytically in section 6.1 and empirically in section 7.3, is that the inferential value of acting on this decomposition is conditional rather than universal. The identity shows that the one-component estimator is displaced from its pharmacological target only by the components' loading onto the contrast used, and, for a biomarker slope, only by the biomarker's covariance with $PB$ and $TV$. Where those terms are zero, the design-contrast one-component analysis is already unbiased, and the production simulation confirms it: with a biomarker coupled to $BR$ alone, the one-component biomarker-by-treatment estimate is unbiased across the whole $PB$-$TV$ grid and the phase-augmented analysis is dominated. The framework's contribution is therefore best read as a map of when decomposition changes inference, not as a blanket prescription to decompose. It makes the trial-design conditions explicit, identifies the attribution and contaminated-biomarker estimands for which decomposition is needed, and clarifies the target estimand for biomarker validation (the biomarker-by-$BR$ association, not the lumped total).

[59]

**Four open questions bound the framework's current scope: the simulation results are pending, the $\eta$ multipliers require sensitivity analysis, non-monotone natural history is not yet handled, and participant-level recovery inherits the identifiability caveats of individual treatment effects.**

- Quantitative bias-versus-power trade-offs are conjectures until the section 7 simulation is run.
- The $\eta$(phase) expectancy-modulation parameters are modelling assumptions requiring explicit sensitivity analysis.
- Cyclical or biphasic $TV$ trajectories exceed the Gompertz form and require flexible or covariate-augmented specifications.
- Participant-level component recovery rests on separating participant-by-component signal from participant-by-period and participant-by-phase variation, the identifiability caveat Senn raises for individual treatment effects.

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Four open questions remain and warrant acknowledgement. First, the simulation study described in section 7 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the $\eta$(phase) multipliers used to parameterise the expectancy modulation of $PB$ are themselves modelling assumptions, and a sensitivity analysis varying $\eta$ across plausible values is mandatory before any clinical conclusion is drawn from the framework's outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic $TV$ patterns require either flexible $TV$ specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that[29] and[28] raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

[60]

**Two companion documents extend the manuscript: a pedagogical narrative and a Gompertz-family sensitivity evaluation.**

- The pedagogy document at `docs/24-component-decomposition-pedagogy.md` provides worked examples for first-time readers.
- The Gompertz evaluation at `analysis/report/07-gompertz-evaluation/`

quantifies trajectory-specification sensitivity.
- Together they address the intuitive and parametric dimensions of the framework that the present manuscript treats briefly.

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The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

[61]

**The components exist in any longitudinal treatment response, but participant-level identifiability requires a within-subject belief-by-exposure contrast that not every placebo-controlled design supplies.**

- The three-component structure is a property of the data-generating process of any repeated-measures treatment study; its identifiability is design-specific.
- Parallel-group placebo control identifies the population-mean $PB$ but not a participant-level $PB$, and supplies no within-subject belief contrast; only designs with such a contrast (blinded or open-hidden discontinuation, randomised withdrawal, crossover, hybrid N-of-1) identify the participant-level decomposition the biomarker question needs.
- The broad claim is therefore scoped to designs supplying a within-subject belief-by-exposure contrast, not to placebo-controlled longitudinal trials in general.

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A final point concerns the scope of the decomposition, where it is essential to distinguish the existence of the components from their identifiability. The claim that an observed longitudinal treatment response is the sum of a pharmacological, a belief-driven, and a natural-history component is a statement about the data-generating process of any repeated-measures treatment study, and in that sense the decomposition is not specific to N-of-1 trials. Identifiability, by contrast, is design-specific, and the distinction between population-level and participant-level recovery is decisive for the biomarker application. A standard parallel-group placebo-controlled trial supplies a between-arm placebo contrast that identifies the population-mean belief response but not a participant-level $PB$, and it contains no within-subject contrast in which belief varies independently of pharmacology; the participant-level biomarker-by-$BR$ slope that the precision-medicine question targets is therefore not identified from such a design, no matter how large it is. The designs that do supply the required within-subject belief contrast are the blinded or open-hidden discontinuation and randomised-withdrawal designs[25,26], the crossover with on/off periods, and the hybrid N-of-1 design developed here; the pharmacometric disease-progression-plus-placebo models from which the framework descends[1,2,34] operate at the group level and recover population-mean components only. We therefore advocate the decomposition as a default lens for longitudinal designs that supply a within-subject belief-by-exposure contrast, with the aggregated N-of-1 case the most demanding instance because it must recover the components from a single participant's trajectory. We do not claim participant-level identifiability from parallel-group placebo control alone, and where only a between-arm placebo contrast is available the framework's claims are restricted to the population mean.

# 10 Conclusions

[62]

**Treatment response in aggregated N-of-1 trials is the sum of three causally distinct components, and the bias from conflating them is governed by the estimand and the biomarker's coupling to the components, not by their raw magnitude.**

- The three components are $BR$ (pharmacological), $PB$ (placebo-belief), and $TV$ (natural history).
- Each has its own biological mechanism, identifiability conditions, and clinical interpretation.
- The lumped pre-post treatment effect is biased by the $PB$ and $TV$ contributions; the biomarker-by-treatment slope is biased only by the biomarker's covariance with $PB$ and $TV$, as Study A confirms.

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In conclusion, three points are to be emphasised. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action ($BR$), placebo-belief response ($PB$), and time-variant natural history ($TV$). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation. Whether conflating them biases a given analysis is, however, governed by the omitted-variable-bias identity of section 6.1, not by the raw magnitude of $PB$ and $TV$: the lumped pre-post treatment effect is displaced by the $PB$ and $TV$ contributions, whereas the biomarker-by-treatment slope is displaced only by the biomarker's covariance with those components. Study A bears this out: varying $PB$ and $TV$ magnitude with a biomarker coupled to $BR$ alone left the one-component estimate unbiased across the entire grid.

[63]

**What matters is the design contrast, not the parametric model: the trial must supply the drug-on/off and belief contrasts, but adding parametric component terms is not free and**

**is not always worth it.**

- Unique component identification requires phase contrasts built into the trial design; the one-component analysis already exploits the drug-on/off contrast.
- Under an uncontaminated biomarker, the phase-augmented analysis adds no bias reduction and loses power, so it is not recommended as a default.
- The phase-augmented and full component-matched analyses are warranted for attribution estimands and for biomarkers that may correlate with $PB$ or $TV$, where the extra structure does work.

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Second, the operative requirement is the design contrast, not the parametric analysis model. The trial design must include the drug-on/off and belief contrasts that identify the components, and the one-component analysis already exploits the drug-on/off contrast through its $D_{bc}$ term. Adding parametric component structure on top is not free: under an uncontaminated biomarker the phase-augmented analysis reduced no bias and lost substantial power, so it should not be adopted as a default. Its value, and that of the full component-matched fit, is confined to the estimands where the extra structure does work, namely the attribution of response to pharmacology versus belief and the analysis of biomarkers that may themselves correlate with $PB$ or $TV$.

[64]

**For predictive-biomarker validation the decomposition matters precisely when the candidate biomarker may correlate with $PB$ or $TV$; this is the case the decisive simulation still has to probe.**

- A biomarker that predicts the lumped total is a pharmacological predictor only if it is uncorrelated with $PB$ and $TV$; otherwise the lumped slope is contaminated.

- Regulatory and prescribing decisions therefore require separating $BR$ from $PB$ and $TV$ whenever biomarker-component coupling cannot be ruled out.
- Study A established the uncontaminated baseline; the contaminated-biomarker cell is the planned empirical test of the failure mode.

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Third, for predictive-biomarker validation the decomposition earns its place precisely when the candidate biomarker may correlate with $PB$ or $TV$. A biomarker that predicts the lumped total is a pharmacological predictor only when it is uncorrelated with the non-pharmacological components; where that cannot be assumed, the lumped slope mixes pharmacological and non-pharmacological prediction, and the regulatory and prescribing decisions that motivate the trial require the separation the decomposition supplies. The coupling-arm study demonstrates this failure directly, and Study C establishes that the bias is recoverable: a belief-covariate decomposition returns the unbiased pharmacological slope across the feasible contamination range, but only on a balanced-placebo design that decouples drug exposure from belief, not on the standard hybrid design. The decisive practical lesson is therefore as much about design as about analysis.

# 11 Conflict of interest

The authors declare no conflicts of interest.

# 12 Funding

This work received no specific funding.

# 13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

# 14 Reproducibility

[65]

**The manuscript is rendered inside a Docker container with locked package versions; the full software environment is pinned by `Dockerfile` and `renv.lock`.**

- R 4.4.x in the zzcollab Docker container provides the execution environment.
- Simulation packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.
- The manuscript build uses `rmarkdown` and `bookdown`.

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This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[66]

**Reproducibility is ensured by a master seed of 20260507 with per-replicate seeds derived by index addition.**

- `set.seed(20260507)` is called at the top of each replicate loop.
- `parallel::mc.set.seed` propagates seeds inside `mclapply`.
- Per-replicate seeds are `master_seed + rep_idx`.

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Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

[67]

**All simulation outputs, driver scripts, pre-registration, and manuscript source files are versioned at documented paths within the repository.**

- Per-replicate outputs and Morris summaries are at `analysis/data/quick-sim/`.
- Driver scripts and the ADEMP pre-registration are at `analysis/scripts/component-decomposition/`.
- Manuscript source, bibliography, and CSL file are at `analysis/report/06-component-decomposition/`.

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Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds`; Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`. Driver scripts are under `analysis/scripts/component-decomposition/`. The pre-registered ADEMP plan is at `analysis/scripts/component-decomposition/00-ademp` The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/06-component-decomposition/`.

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